

SAURABH SALVE

AI Backend Engineer | FastAPI & LLM Systems | Generative AI

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SUMMARY

Backend AI Engineer with hands-on experience building scalable FastAPI services, retrieval-augmented generation (RAG) systems, and agentic AI workflows. Strong foundation in Python backend development, asynchronous APIs, and LLM orchestration, complemented by merged open-source contributions to FastAPI and current contract engineering work with AfterQuery Experts.

SKILLS

Languages: Python, JavaScript, SQL

Backend: FastAPI, REST APIs, AsyncIO, Background Tasks, Streaming (SSE), JWT Authentication, Rate Limiting

AI / LLM: LangChain, LangGraph, OpenAI API, Gemini API, RAG, FAISS (Vector Search), Prompt Engineering, Multi-Agent Orchestration

Databases: PostgreSQL, Supabase, Redis (Caching)

Cloud: AWS (EC2, Lambda), GCP (Vertex AI)

DevOps: Docker, CI/CD, Git, Logging

Frontend: React, Next.js, Streamlit

PROJECTS

Repo2Viral — Async FastAPI Service for Repo-to-Content Generation | [GitHub](#) | [Live](#)

- Converts a GitHub repository into documentation and short-form video content; the API queues generation as a background task and returns immediately instead of blocking on the LLM call.
- Chunks source code with syntax-aware (AST-based) splitting rather than naive text splitting, so functions and classes stay intact; routes code-understanding calls to GPT-4 and content-generation calls to GPT-3.5-turbo; persists job state in Supabase/PostgreSQL behind a Next.js frontend that polls job status.

Tech: FastAPI • PostgreSQL (Supabase) • OpenAI API • Next.js • Docker • Redis

Math Mentor — Multimodal Math Tutor | [GitHub](#)

- Solves handwritten JEE-level math problems by combining OCR, LLM reasoning, and symbolic verification.
- Routes handwritten input through a 3-tier OCR fallback (EasyOCR, then GPT-4o Vision, then Mistral OCR as the primary path after benchmarking accuracy on handwritten notation), checks each solution with two independent SymPy verification passes inside a 6-agent LangGraph pipeline, and retrieves reference material through a LangChain + FAISS RAG layer over OpenAI embeddings.

Tech: LangGraph • LangChain • FAISS • OpenAI API • SymPy

Shadow Board — Multi-Agent Decision Simulation | [GitHub](#)

- Runs simulated board debates (Legal, Finance, Tech, Healthcare, Retail) between LLM agents and compiles the output into a written strategy brief.
- Fixed a state-collision bug in parallel agent execution by streaming each agent's output into an isolated buffer per agent, merging only at the synthesis step; added human-intervention hooks, Slack notifications, and Supabase-based auth/persistence.

Tech: Gemini API • LangChain • Supabase (Auth + PostgreSQL) • Slack API

OPEN SOURCE — FASTAPI (FASTAPI/FASTAPI)

- Fixed a bug where the declared status_code was ignored on SSE and JSONL streaming responses, causing a mismatch between the generated OpenAPI schema and actual runtime behavior — merged ([PR #15937](#)).
- Added the missing API reference documentation page for the fastapi.sse module — merged ([PR #15930](#)).

EXPERIENCE

Software Engineer (Contract) — AfterQuery Experts (Y Combinator-backed)

Aug 2026 – Present

- Resolve real-world engineering issues in unfamiliar, production-style GitHub repositories — debugging, implementing fixes, and writing regression tests to validate expected behavior.
- Own the full engineering workflow on Python/FastAPI codebases: analyzing requirements, implementing changes, running test suites, and preparing Git patches/PRs for review.
- Author evaluation-ready test suites and task specifications for SWE-bench-style AI coding-agent evaluations, sharpening codebase exploration and root-cause analysis skills.

Data Science Intern — IBM SkillsBuild

Jun 2024 – Aug 2024

- Manual churn analysis on 100K+ customer records was slow to update; built a feature-engineered classification pipeline in Python, Pandas, and Scikit-learn, and replaced manual reporting with a Streamlit dashboard.
- Reached 83% model accuracy and cut analysis turnaround by 40%; the dashboard was adopted by business teams.

Cloud Computing Intern — Amazon Web Services

Jun 2023 – Aug 2023

- Application releases were manual and did not scale under load; containerized the app on AWS EC2 with Docker auto-scaling and built a serverless ETL pipeline (Lambda + CodePipeline) with full CI/CD.
- Reduced peak-load latency by 30% and removed manual release steps while processing 10,000+ daily events.

ACHIEVEMENTS

- National Finalist, Google GenAI Exchange Hackathon — top 5% nationwide for an end-to-end agentic LLM solution built from scratch.
- Google Cloud Generative AI Academy — certification.

EDUCATION

B.Tech, Computer Science Engineering — Deogiri Institute of Engineering and Management Studies, Aurangabad

2022 – 2026